

A Statistical Assessment of the Impact of Machine Learning Inputs on the Performance of Integer-Valued Markowitz-Type Portfolios

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ABSTRACT

The dynamic and volatile nature of financial markets requires decision tools that combine accurate stock movement forecasts with optimal portfolios derived from formulations that incorporate real-world trading constraints. This paper presents an integer-valued portfolio optimization framework that integrates next-day stock closing price predictions from machine learning (ML) models (XGBoost) into several Markowitz-type formulations. We evaluate multiple formulations including diversification constraints, explicit short-selling models (standard and volatility-normalized / “Black–Scholes”-style), a Meta-Markowitz ensemble that hedges strategic risk across portfolios themselves, and a robust optimization variant under box uncertainty. Experiments on the Indian SENSEX constituent stocks (2010–2024 training window) show that ML-informed integer portfolios outperform the SENSEX and Markowitz-without-ML benchmarks with respect to five portfolio performance metrics including the Sharpe and Sortino ratios, while robust and meta strategies reduce downside risk. These experiments involve a comprehensive statistical analysis pipeline: Ljung-Box and augmented Dickey-Fuller tests for autocorrelation and stationarity prior to statistical comparison; Shapiro-Wilk tests of normality to determine whether nonparametric or parametric comparisons must be performed; and finally, a variety of parametric (Student’s t) and nonparametric (Friedman, Wilcoxon signed-rank) tests to assess performance of each ML-informed portfolio against the benchmarks.

1. Introduction

Markowitz in 1952 proposed the mean-variance model for optimal portfolio selection (Ruppert, 2004). Many variations on this approach have since been developed (Engels, 2004), and in recent times, the incorporation of inputs generated from machine learning (ML) methods within Markowitz-based portfolio optimization models has also been considered (Brogaard and Zareei, 2023; Ma, Han and Wang, 2021). However, in our knowledge, there has been no published study that performs a comprehensive statistical assessment of whether ML inputs indeed improve the performance of Markowitz-based optimal portfolios.

We recall that the classical Markowitz portfolio optimization model is of the form:

$$\begin{aligned} \max_w \quad & w^T \mu - \gamma w^T \Sigma w \\ \text{s.t.} \quad & w^T \mathbf{1} = 1 \\ & w \geq 0 \end{aligned} \tag{1}$$

In equation 1, $w \in \mathbb{R}^{n+} \cup \{0\}$ represents the portfolio allocation (with each component w_i being the fraction of the budget allocated to a stock). μ and Σ represent the mean returns vector and covariance matrix of the returns, and $\gamma > 0$ is the risk aversion parameter (higher values indicate lower risk tolerance and vice versa). In this paper, we consider the integer-valued or ‘round-lot’ version of the classical Markowitz model, which allows analysts to directly compute the number of shares of each stock to be purchased. This model is given below.

ORCID(s):

$$\begin{aligned}
& \underset{w}{\text{maximize}} && w^T \mu - \gamma w^T \Sigma w \\
& \text{subject to:} && \mathbf{P}^T w \leq \mathbf{B} \\
& && w_i \geq 0, w_i \in \mathbb{Z}, \quad i = 1, \dots, 53
\end{aligned} \tag{2}$$

In equation 2, $\mathbf{P}$ is the price vector of the stocks under consideration, and $\mathbf{B}$ is the total budget available for investment. In this work, we consider, in addition to the model in equation 2, its variants that (a) ensure diversification of the portfolio, (b) allow short-selling and (c) include Black-Scholes inspired stock price normalization by its volatility. We use these, along with a stock market-weighted index (the Indian SENSEX index) based portfolio, as benchmarks for our ML-based portfolios. ML model outputs are incorporated in the following ways in the above variants of the Markowitz model: (a) instead of using average stock prices / returns (i.e., μ), we use ML predictions of stock prices, and (b) we include a function of ML model prediction accuracy for each stock as a measure of its risk. We also develop an approach to hedge risk across portfolios, called the ‘meta-Markowitz’ model. We consider daily portfolio reallocations, and use the XGBoost ML model to generate the next-day stock closing price predictions.

We then conduct a comprehensive statistical assessment of the performance of each ML-based Markowitz portfolio against the benchmarks, wherein the performance of each portfolio is measured across five different metrics, including the Sharpe and Sortino ratios. The statistical testing pipeline includes assessments of autocorrelation and stationarity of each portfolio’s returns across the test period, tests of normality to determine the appropriate test procedure to be used to assess performance against the benchmarks, and the performance assessment tests themselves.

We briefly discuss the relevant literature and research contributions of our work below.

The literature demonstrates a decisive shift from linear econometric models (e.g., ARIMA) to non-linear ML techniques for return prediction. Support vector regression (SVR) has emerged as a robust tool for financial forecasting due to its ability to model non-linear relationships. Paiva, Cardoso, Hanaoka and Duarte (2019) proposed a fusion approach using SVM for stock selection and SVR for return prediction in the Brazilian market, highlighting the impact of brokerage costs on net returns. In the Indian context, Mishra and Padhy (2019) validated that SVR with Gaussian kernels significantly reduces Mean Absolute Percentage Error (MAPE) compared to traditional regression. Furthermore, Mishra, Padhy, Mishra and Misra (2021) introduced a "LASSO-TLBO-SVR" hybrid model, utilizing meta-heuristics to optimize hyperparameters, thereby enhancing the model’s predictive power.

Deep learning architectures have further pushed the methodological frontier. Ma et al. (2021) demonstrated that Deep Belief Networks (DBN) capture volatility clusters better than ARIMA models. Sharma and Shekhawat (2022) developed a hybrid DBRNN (Deep Belief + Recurrent Neural Network) optimized via the Harris-Hawks algorithm, specifically testing on BSE data. To address model variance, Chen, Zhang, Mehlaawat and Jia (2021) and Zhou, Song, Ren and Yu (2022) utilized ensemble methods like XGBoost, finding that aggregating weak learners significantly improves the Sharpe ratio and robustness of the predictions.

While ML improves point estimates of returns (μ), it does not address errors in the covariance matrix (Σ). Rosadi, Setiawan, Templ and Filzmoser (2020) integrated robust statistics, specifically Minimum Covariance Determinant (MCD) estimators, into the Mean-Variance framework. Their findings indicate that robust estimators, when combined with integer constraints, drastically reduce portfolio turnover and weight instability. Similarly, Xidonas, Steuer and Hassapis (2020) provide a comprehensive classification of robust optimization, noting that while RO handles uncertainty effectively, it can be overly conservative without accurate predictive inputs.

A significant disconnect exists between theoretical continuous models and the discrete reality of trading.

- **Cardinality:** Chen et al. (2021) empirically determined that a portfolio size of roughly 10 stocks offers optimal diversification benefits, justifying the use of cardinality constraints ($\sum z_i = K$).
- **Transaction lots:** Rosadi et al. (2020) explicitly formulated the portfolio problem using integer transaction lots, avoiding the "odd lot" costs associated with rounding continuous weights.
- **Indian Market Specifics:** Most reviewed studies (Ma et al., 2021; Zhou et al., 2022) utilize simplified transaction cost models. There is a lack of research incorporating the specific short-selling costs and platform fees inherent to the Indian Securities Lending and Borrowing (SLB) mechanism.

The above discussion reveals that while ML, robust optimization, and integer programming have advanced individually in the context of Markowitz-style portfolio optimization, few studies unify them. For example, Rosadi et al.

(2020) handles integer lots and robust covariance but lacks ML predictions. Conversely, Sharma and Shekhawat (2022) uses advanced ML but ignores integer constraints.

In summary, the contributions of our work include:

- Several Markowitz-style integer formulations that incorporate ML predictions in mean/expected-return estimates and in risk adjustments.
- New formulations to explicitly model short-selling in the Indian market context (including a volatility-normalized variant).
- A *meta-Markowitz* ensemble that treats strategy returns as assets and optimizes weights across strategies.
- A robust Markowitz formulation under box uncertainty to account for estimation risk.
- Thorough experimental evaluation and statistical significance testing against the SENSEX benchmark.

We now describe the stock price prediction methodology and the portfolio optimization models that we develop in this article.

2. Methodology

In this paper, we develop a framework for a comprehensive statistical assessment of whether machine learning based predictions of stock prices can improve the performance of Markowitz-based portfolio optimization strategies. In the context of a single portfolio optimization formulation, the framework broadly involves the following stages.

- 1 First, for each stock considered, we generate an ML-based (for the ML-based Markowitz portfolio) prediction of its price for the time period under consideration (e.g., its close price $\hat{C}_t$ on investment day t).
- 2 The price predictions for each stock are input into the Markowitz-based portfolio optimization formulation under consideration, which is then solved to generate the numbers of each stock to be purchased. This is then used, along with the actual stock prices c_t , to compute the portfolio return $r_{ML(t)}$ for day t .
- 3 Repeat steps 1 and 2 for each time unit in the testing period of length L (e.g., for each day t in a test period comprising 360 days for which actually realized stock prices are available) to obtain a vector of returns $R_{ML} = (r_{ML(1)}, r_{ML(2)}, \dots, r_{ML(L)})^T$ for the ML-based portfolio across the test period L .
- 4 Repeat steps 1 through 3 for a benchmark portfolio that *does not* use ML predictions of stock price to generate a vector of returns $R_B = (r_{B(1)}, r_{B(2)}, \dots, r_{B(L)})^T$. For example, the original Markowitz portfolio optimization formulation can be used as a benchmark
- 5 Conduct the appropriate statistical tests using R_{ML} and R_B at a prespecified level of significance to assess whether there is a statistically significant difference between performance of the ML-based and benchmark portfolios.

In this work, in order to conduct a more comprehensive assessment of whether ML inputs improve Markowitz-based portfolio performance, we apply the above procedure (steps 1 through 5) for multiple Markowitz-based portfolios that utilize ML inputs, and conduct the statistical assessment for not just one portfolio performance metric (such as the mean returns), but for five portfolio performance metrics. Our statistical assessment methodology in itself can be broken down into the following steps, and a key contribution of this study involves the specification of this statistical assessment roadmap for integer-valued Markowitz-based portfolio optimization.

- 1 Check whether R_{ML} and R_B can be considered to be independent and identically distributed (*iid*), via tests of autocorrelation and stationarity.
- 2 Check whether R_{ML} and R_B follow the Gaussian distribution, via tests of normality, so that parametric or nonparametric test procedures can accordingly be selected for assessing whether there are statistically significant differences between the mean returns from both portfolio optimization approaches.

3 Conduct the appropriate test.

The above procedure is applied to assess whether there is a statistically significant difference between the mean returns from the ML-based portfolio and the benchmark portfolio. However, to conduct statistical analyses around the other metrics, for which only one realization is obtained from each vector of returns (R_{ML} and R_B). Therefore, to assess whether there are statistically significant differences between the ML-based portfolio and the benchmark portfolios as measured by other metrics (such as the Sharpe ratio or the Sortino ratio), R_{ML} and R_B are decomposed into, say, M smaller samples, from which M realizations of the metric in question can be computed. The same procedure can then be applied to assess, statistically, the performance of the ML-based portfolio against the benchmark portfolio in terms of the metric in question (e.g., the Sharpe ratio).

We now describe in detail each component of the above framework. We begin with the ML-based generation of stock price predictions.

2.1. Stock Price Prediction

We consider a relatively short-term portfolio optimization problem for the empirical work in this study; however, the approach described in the paper can be applied to any trading window assuming that sufficient data for the statistical analysis can be generated.

We assume that the investor allocates and reallocates their portfolio on a daily basis - i.e., a one-day trading window is assumed. Specifically, we assume that the (re)allocation of the portfolio is made either at the end of day $t - 1$ or at the beginning of day t , and that returns are realized at the end of day t . Therefore, for each stock, we utilize data for day $t - 1$, including the close price of the stock on day $t - 1$, to predict the close price on day t .

Algorithm 1 Walk-Forward ML Model Training

Require:

- S_{all} : Set of all stocks to be modeled (the 53 SENSEX stocks in this scenario)
- D_s : Complete historical feature/target dataset for a stock $s \in S_{all}$
- Y_{start} : First year of training data (e.g., 2010)
- Y_{test_start} : First year of the test period (e.g., 2023)
- Y_{test_end} : Final year of the test period (e.g., 2024)
- M_{base} : Base model (XGBoost Regressor)
- $T_{optimizer}$: Hyperparameter tuning method (Optuna here)
- P_{best} : Best set of hyperparameters found (Optuna here)
- L_{metric} : Loss metric for tuning (RMSE here)
- C : represents the set of Constraints

Ensure:

$\mathcal{M}$: A collection of trained models, $M_{s, y_{test}}$, for each stock s and test year y_{test}

```

1: procedure TRAINMODELS( $S_{all}$ ,  $D$ ,  $Y_{start}$ ,  $Y_{test\_start}$ ,  $Y_{test\_end}$ ,  $M_{base}$ ,  $T_{optimizer}$ )
2:    $\mathcal{M} \leftarrow \text{EmptyMap}$ 
3:   for all  $s \in S_{all}$  do
4:     for all  $y_{test}$  from  $Y_{test\_start}$  to  $Y_{test\_end}$  do
5:        $y_{train\_end} \leftarrow y_{test} - 1$ 
6:        $D_{train} \leftarrow \{(x, y) \in D_s \mid \text{year}(x) \geq Y_{start} \text{ AND } \text{year}(x) \leq y_{train\_end}\}$ 
7:        $(X_{train}, Y_{train}) \leftarrow D_{train}$ 
8:        $P_{best} \leftarrow T_{optimizer}(M_{base}, X_{train}, Y_{train}, \text{metric} = L_{metric})$ 
9:        $M_{final} \leftarrow \text{train}(M_{base}, X_{train}, Y_{train}, P_{best})$ 
10:       $\mathcal{M}[s, y_{test}] \leftarrow M_{final}$ 
11:     end for
12:   end for
13:   return  $\mathcal{M}$ 
14: end procedure

```

The empirical analysis is based on SENSEX constituent data spanning from 1 January 2010 to 1 January 2022 excluding 2020 due to exceptional behavior due to Covid-19. Our dataset comprises 53 (see Table 4) financially

Algorithm 2 Daily Walk-Forward Portfolio Optimization Backtest**Require:**

$\mathcal{M}$: The set of trained models from Algorithm 1
 D_{all} : Complete feature and price data for all $s \in S_{\text{all}}$
 T_{start} : First day of the backtest (e.g., '2023-01-01')
 T_{end} : Last day of the backtest (e.g., '2024-12-31')
 F_{set} : Set of all portfolio formulations (Type 1 & Type 2 with variations)
 B : Daily Budget (INR 1,00,000 here)
 γ : Risk Aversion parameter
 L_{cov} : Lookback period for covariance matrix (60 days here)
 a_i : Complement of MAPE vector, $a_i(i) = 1 - \text{MAPE}(i)$ for stock i

Ensure:

R_{daily} : Collection of daily returns and portfolio weights ($w_{t,f}$) for each $f \in F_{\text{set}}$
1: **procedure** RUNBACKTEST($\mathcal{M}, D_{\text{all}}, T_{\text{start}}, T_{\text{end}}, F_{\text{set}}, B, \gamma, L_{\text{cov}}$)
2: Initialize $R_{\text{daily}} \leftarrow \text{EmptyMap}$ for each $f \in F_{\text{set}}$
3: **for all** t **from** T_{start} **to** T_{end} **do**
4: $y_{\text{test}} \leftarrow \text{year}(t)$
5: $\hat{C}_t \leftarrow \text{EmptyVector}$
6: **for all** $s \in S_{\text{all}}$ **do**
7: $M \leftarrow \mathcal{M}[s, y_{\text{test}}]$
8: $x_{t,s} \leftarrow \text{GetFeatures}(D_{\text{all}}, s, t)$
9: $\hat{c}_{t,s} \leftarrow M.\text{predict}(x_{t,s})$
10: Append $\hat{c}_{t,s}$ to $\hat{C}_t$
11: **end for**
12: $C_{t-1} \leftarrow \text{GetActualClosePrices}(D_{\text{all}}, t-1)$
13: $P_t \leftarrow \text{GetInvestmentPrices}(D_{\text{all}}, t)$
14: $\Sigma_t \leftarrow \text{CalculateCovariance}(D_{\text{all}}, \text{from } t - L_{\text{cov}} \text{ to } t-1)$
15: $a_i \leftarrow 1 - \text{MAPE}(i)$ for each stock i
16: $\sigma \leftarrow \text{CalculateVolatility}(D_{\text{all}}, \text{from } t - L_{\text{cov}} \text{ to } t-1)$
17: **for all** $f \in F_{\text{set}}$ **do**
18: Define μ based on formulation : type-I or type-II:
19: Solve $w_{t,f} \leftarrow \arg \max_w \text{Objective}(f)$ subject to budget, diversification and integer constraint type C
20: $C_t \leftarrow \text{GetActualClosePrices}(D_{\text{all}}, t)$
21: $\text{Return}_{t,f} \leftarrow w_{t,f}^T (C_t - P_t)$
22: Append $(\text{Return}_{t,f}, w_{t,f})$ to $R_{\text{daily}}[f]$
23: **end for**
24: **end for**
25: **return** R_{daily}
26: **end procedure**

sound and well-established companies that are listed on the Bombay Stock Exchange (BSE). Data were obtained using Quandl's Python package and the Yahoo Finance API (yfinance), supplemented by data from the BSE SENSEX index and NYSE Standard & Poor's 500 index. Table 1 summarizes the ML model training specifications.

We consider the XGBoost model here in our work. While the primary purpose of our work is to compare the effect of incorporating ML into classical Markowitz formulation via rigorous statistical testing, and not to find the best ML model for stock price prediction, we note that the XGBoost model has consistently found to be highly successful in working with tabular data (Shwartz-Ziv and Armon, 2022). Hence we chose this ML method for predicting stock prices for our framework. The XGBoost ML model is trained on 19 features consisting of historical price data, technical indicators, and market indices. These were preprocessed as per the model requirement. The feature set is described in Table 2.

The XGBoost ML model is a parallelized, scalable and computationally optimized version of the gradient boosted decision trees ensemble classification and regression model. Gradient boosted decision trees are an ensemble of weak learners (i.e., individual decision trees). This ensemble decision tree model differs from other such models such as the random forest ensemble decision tree model in the way weak learners are iteratively (additively) added in a manner

Algorithm 3 Meta-Markowitz Strategy Aggregation**Require:**

R_{daily} : Daily return streams for the 11 base formulations (from Algorithm 2)
 $T_{\text{start}}, T_{\text{end}}$: The backtest date range
 L_{meta} : Lookback period for meta-parameters (60 days)
 γ_{meta} : Risk aversion parameter for the meta-portfolio

Ensure:

R_{meta} : A time-series of daily returns for the aggregated Meta-Markowitz portfolio

```

1: procedure RUNMETAMARKOWITZ( $R_{\text{daily}}, T_{\text{start}}, T_{\text{end}}, L_{\text{meta}}, \gamma_{\text{meta}}$ )
2:    $R_{\text{meta}} \leftarrow \text{EmptyList}$ 
3:    $F_{\text{base}} \leftarrow \text{GetBaseFormulations}(R_{\text{daily}})$ 
4:   for all  $t$  from  $(T_{\text{start}} + L_{\text{meta}})$  to  $T_{\text{end}}$  do
5:      $D_{\text{past}} \leftarrow \text{GetReturns}(R_{\text{daily}}, F_{\text{base}}, \text{from } t - L_{\text{meta}} \text{ to } t - 1)$ 
6:      $\mu_{\text{meta}} \leftarrow \text{Average}(D_{\text{past}})$ 
7:      $\Sigma_{\text{meta}} \leftarrow \text{Covariance}(D_{\text{past}})$ 
8:      $w_{\text{meta}} \leftarrow \arg \max (w^T \mu_{\text{meta}} - \gamma_{\text{meta}} w^T \Sigma_{\text{meta}} w)$ 
       subject to:  $\sum_{i=1}^{11} w_i \leq 1, w_i \geq 0$ 
9:      $R_t \leftarrow \text{GetReturns}(R_{\text{daily}}, F_{\text{base}}, \text{at day } t)$ 
10:     $\text{Return}_{t,\text{meta}} \leftarrow w_{\text{meta}}^T \cdot R_t$ 
11:    Append  $\text{Return}_{t,\text{meta}}$  to  $R_{\text{meta}}$ 
12:  end for
13:  return  $R_{\text{meta}}$ 
14: end procedure

```

Table 1

Dataset Organization and Machine Learning Model Training

Feature	Description
Data Period	SENSEX data candles from 1st January 2010 to 1st January 2022 including open, close, low, high, turnover, contracts, and volume.
Data Source	Data downloaded using Quandl's Python package and Yahoo Finance API (yfinance); additional data from the BSE SENSEX index and NYSE Standard & Poor's 500 index.
Technical Indicators	15 different technical indicators were generated from the candlestick data.
Dataset Features	25 features including technical indicators were used.
Dataset Samples	Total of 3,473 samples, approximately equally balanced across all stocks.
Dataset Split	Split in an 80:20 ratio for training and validation, further divided into 75:25 for hyperparameter tuning.

equivalent to the implementation of gradient descent for optimal parameter estimation for ML models. Gradient boosted trees models are constructed by initializing the loss function with a single weak learner (e.g., a single decision tree), and then adding another weak learner. The weights of the newly added weak learner are then optimized via a loss function defined across the training set. The optimization of the weights of the newly added weak learner is performed using a second-order Taylor series expansion of the loss function, using first- and second-order derivative information, and hence the name gradient boosted decision trees. A representation of the loss function $L_{\text{sub}}(t)$ when the t th weak learner is added is provided below.

$$L_{(t)} = \sum_{i=1}^n \left[L_{(t-1)} + D_1 W_{t(i)} + D_2 W_{t(i)}^2 \right] + R_t + \text{constant} \quad (2)$$

Table 2

Feature Set of XGBoost Model Training.

Features	Remarks	Time Periods
Open	Price at which stock is trading when market opens	Last Day
Low	Lowest price at which stock trades during the day	Last Day
High	Highest price at which stock trades during the day	Last Day
Volume	Total number of shares that have traded during the day	Last Day
After Technical Analysis on Raw Data		
Simple Moving Averages	Close Price Moving Average over a period	50 & 200 day period
Exponential Moving Averages	Close Price Moving Average with more weight on recent data	50 & 200 day period
Relative Strength Index	Indicator measuring the magnitude of recent price changes. Evaluates overbought or oversold conditions for a stock	14 day period
Moving Average Convergence Divergence	Trend-following momentum indicator. Shows the relationship between two moving averages	Fast Period=12, Slow Period=26, Signal Line=9
Bollinger Bands	Indicator defined by a set of lines plotted two standard deviation away from a SMA of the close prices	20 day period
Average True Range	Measures close, price volatility	14 day period
On-Balance Volume	Adds/subtracts daily volume depending on price movement, indicating cumulative pressure	Cumulative
Commodity Channel Index	Indicator used to assess price trend direction and strength	14 day period
Stochastic Oscillator	Indicates the location of the close price relative to the high-low price range over a fixed number of periods	Fast Period=14, Slow Period=3
Parabolic Stop and Reverse	Indicates the possible points of reversal	Acceleration=0.02
Chaikin Money Flow	Measures the accumulation-distribution line of the MACD Line	20 day period
Williams %R	Indicates closeness of Close price relative to High of trading period	14 day period
Ultimate Oscillator	Combines momentum indicator over 3 different time-periods	7, 14, 28 day period
Accumulation Distribution Line	Cumulative Money Flow Indicator. Flow of money into and out	Cumulative

In the above equation, D_1 and D_2 represent the first- and second-order derivatives of the loss function at the $(t - 1)^{\text{th}}$ step; $W_{t(i)}$ represents the t^{th} weak learner's prediction for the i^{th} sample; and R_t represents the regularization term controlling the complexity of the t^{th} weak learner.

The organization of the training set for the XGBoost model is straightforward: each sample consists of a feature vector of dimension 25, and the target is the closing price of a trading day. The number of such samples is 3,473.

To make the predictions more realistic, we used an incremental training approach. For the test year 2023, each machine learning model was trained on data available up to 2022. Likewise, for 2024 predictions, the models were retrained using all data up to 2023. This setup mirrors how models would be updated in real-world applications, where new information becomes available each year and is incorporated before making future forecasts.

Feature importance emphasized lagged price, moving averages, and RSI-based features. Hyperparameter tuning (Table 3) employed Optuna with RMSE as the objective. Model performance metrics (RMSE, MAE, MAPE, R^2) are reported in Table 4.

Table 3

Hyperparameter search space for the XGBoost model

Hyperparameter	Search Space
Number of estimators	{100, 200, 300}
Maximum depth	{10, 20, 30}
Learning rate	{0.001, 0.005, 0.0015}
Gamma	{0.001, 0.005}

2.2. Portfolio Optimization Formulations

The original Markowitz mean-variance portfolio optimization formulation (equation 1) utilizes the mean and variance of the returns of each stock, estimated using historical data, considered for allocation. The integer-valued version formulation is given in equation 2.

Here μ and Σ represent the mean returns vector and covariance matrix of returns respectively of the $n = 53$ stocks considered for allocation. μ and σ are estimated using the historical data. w is the vector of decision variables, and it represents the number of shares purchased of each stock, and γ represents the risk-aversion parameter (higher values represent greater degrees of risk aversion). $\mathbf{P}$ is the investment day price vector and $\mathbf{B}$ is defined as the Budget variable.

Note that the investment budget value adopted in our computational experiments is INR 1,00,000, and also that the optimization problem in equation 1 is convex.

In this study, we consider three variants of the classical Markowitz mean-variance portfolio optimization model besides the other formulations which we will introduce next. The baseline model (formulation 2), denoted as **MKW-N**, represents the original Markowitz formulation. The second variant, **MKW-S**, incorporates short-selling constraints, allowing portfolio weights to be negative.¹ The third variant, **MKW-D**, introduces a diversification constraint to ensure a balanced allocation of weights across assets. The parameters μ and Σ can be estimated using historical data as follows:

$$\mu_i = \frac{1}{60} \sum_{k=t-60}^{t-1} r_{i,k}, \quad (3)$$

$$\Sigma_{ij} = \text{Cov}(r_i, r_j). \quad (4)$$

where μ_i represents the average daily returns of stock i over the last 60 trading days, and Σ_{ij} denotes the covariance of past returns between stocks i and j .

Daily returns are computed as $r_t = C_t - C_{t-1}$, where C_t is the close price on day t .

We propose a series of ML-integrated portfolio optimization formulations that progressively put risk management constraints into the classical Markowitz framework. Each formulation builds upon the previous one, addressing specific limitations and incorporating additional dimensions of risk.

2.2.1. ML Formulation Type-1

Our first modification to the classical Markowitz model substitutes the traditional expected return vector with ML-predicted close prices. Specifically, we set $\mu = \hat{C}_t$, where $\hat{C}_t$ represents the vector of ML-predicted close prices for day t . The covariance matrix Σ remains unchanged from the original formulation, computed using historical return data.

This leads to our first ML-enhanced formulation (**type1-standard-normal**) (**ML1-N**):

$$\begin{aligned} & \underset{w}{\text{maximize}} && w^T \hat{C}_t - \gamma w^T \Sigma w \\ & \text{subject to:} && P^T w \leq B \\ & && w_i \geq 0, \quad i = 1, \dots, n = \quad w_i \in \mathbb{Z}, \quad i = 1, \dots, n \end{aligned} \quad (3.2)$$

The objective function maximizes expected returns (based on ML predictions) while penalizing portfolio variance, subject to budget and non-negativity constraints.

¹ Short selling involves borrowing and selling stocks with the expectation of buying them back at a lower price.

While formulation (3.2) incorporates ML predictions into the return component, it treats all predictions with equal confidence. In practice, ML models exhibit varying prediction accuracy across different stocks. A model that predicts stock i with 95% accuracy should intuitively receive more weight than one predicting stock j with only 70% accuracy. The standard formulation fails to account for this heterogeneity in prediction quality.

To address this limitation, we introduce a prediction accuracy term that penalizes investments in stocks with less reliable forecasts. We define the accuracy coefficient for stock i as $a_i(i) = 1 - \text{MAPE}(i)$, where $\text{MAPE}(i)$ is the Mean Absolute Percentage Error of the ML model for stock i . Higher values of $a_i(i)$ indicate more accurate predictions and thus lower prediction risk.

We augment the objective function to include both covariance-based market risk (Σ) and prediction accuracy risk (a_i) (i.e., the *type1-both-normal* formulation).

$$\begin{aligned} & \underset{w}{\text{maximize}} && w^T \hat{C}_t - \gamma_1 w^T a_i - \gamma_2 w^T \Sigma w \\ & \text{subject to:} && P^T w \leq B \\ & && w_i \geq 0, \quad i = 1, \dots, n \\ & && w_i \in \mathbb{Z}, \quad i = 1, \dots, n \end{aligned} \tag{3.3}$$

The term $\gamma_1 w^T a_i$ penalizes allocation to stocks with lower prediction accuracy, while $\gamma_2 w^T \Sigma w$ maintains the traditional volatility penalty. The parameters γ_1 and γ_2 allow the investor to balance between prediction risk and market risk according to their confidence in the ML models versus historical volatility patterns.

Both formulations (3.2) and (3.3) permit highly concentrated portfolios. For instance, if a single stock has both high predicted returns and low risk (or high prediction accuracy), the optimizer may allocate a disproportionate fraction of the budget to that stock. Such concentration exposes the investor to idiosyncratic risk, the possibility that a single stock's adverse movement could devastate the entire portfolio. This violates a fundamental principle of portfolio theory: diversification.

To ensure adequate diversification, we impose upper bounds on individual stock positions:

$$0 \leq w_i \leq 0.5 \times \frac{B}{P_i}, \quad i = 1, \dots, n \tag{3.4}$$

This constraint limits each stock position to at most 50% of the total budget, preventing over-concentration while still allowing meaningful allocation to high-conviction opportunities. The term $\frac{B}{P_i}$ represents the maximum number of shares of stock i that could be purchased with the full budget; thus, the constraint caps each position at half this maximum.

Integrating constraint (3.4) with the standard ML formulation yields the *type1-standard-diversification* formulation:

$$\begin{aligned} & \underset{w}{\text{maximize}} && w^T \hat{C}_t - \gamma w^T \Sigma w \\ & \text{subject to:} && P^T w \leq B \\ & && 0 \leq w_i \leq 0.5 \times \frac{B}{P_i}, \quad i = 1, \dots, n \\ & && w_i \in \mathbb{Z}, \quad i = 1, \dots, n \end{aligned} \tag{3.5}$$

Similarly, our most comprehensive type-1 formulation combines prediction accuracy adjustment with diversification constraints (the *type1-both-diversification* formulation):

$$\begin{aligned}
& \underset{w}{\text{maximize}} && w^T \hat{C}_t - \gamma_1 w^T a_t - \gamma_2 w^T \Sigma w \\
& \text{subject to:} && P^T w \leq B \\
& && 0 \leq w_i \leq 0.5 \times \frac{B}{P_i}, \quad i = 1, \dots, n \\
& && w_i \in \mathbb{Z}, \quad i = 1, \dots, n
\end{aligned} \tag{3.6}$$

By systematically building from the classical Markowitz model to our comprehensive formulation (3.6), we create a robust portfolio optimization framework that leverages ML predictions while explicitly managing multiple dimensions of risk and ensuring prudent diversification.

2.2.2. Type-2 ML-based Formulations

While Type 1 formulations (Section 2.2.1) directly use predicted prices in the objective function, an alternative approach is to construct an expected returns vector from ML predictions. This Type 2 specification more closely mirrors the classical Markowitz framework's use of return expectations rather than absolute price levels.

We define the expected return vector using the difference between predicted and current prices: $\mu = \hat{C}_t - C_{t-1}$, where $\hat{C}_t$ represents the ML-predicted close prices for day t , and C_{t-1} are the actual close prices from the previous trading day. This formulation captures the anticipated price movement, which directly represents the expected profit per share from holding each stock. The advantages of the return-centric formulation are as follows:

- (a) It aligns with traditional portfolio theory by focusing on returns rather than absolute prices.
- (b) Returns across assets become comparable even when their price levels differ significantly.
- (c) The objective function directly represents the expected portfolio profit derived from price changes.

We now develop the Type 2 analogs of our four Type 1 formulations, maintaining the same progressive enhancement structure.

The baseline Type 2 formulation (the *type2-standard-normal* formulation) substitutes the return-based objective while maintaining covariance-based risk:

$$\begin{aligned}
& \underset{w}{\text{maximize}} && w^T (\hat{C}_t - C_{t-1}) - \gamma w^T \Sigma w \\
& \text{subject to:} && P^T w \leq B \\
& && w_i \geq 0, \quad i = 1, \dots, n \\
& && w_i \in \mathbb{Z}, \quad i = 1, \dots, n
\end{aligned} \tag{3.8}$$

Here, the objective $w^T (\hat{C}_t - C_{t-1})$ represents the total expected profit from the portfolio based on predicted price changes, while $\gamma w^T \Sigma w$ penalizes portfolio variance as before.

Analogous to formulation (3.3), we incorporate the prediction accuracy term to account for heterogeneous model performance across stocks (the *type2-both-normal* formulation):

$$\begin{aligned}
& \underset{w}{\text{maximize}} && w^T (\hat{C}_t - C_{t-1}) - \gamma_1 w^T a_t - \gamma_2 w^T \Sigma w \\
& \text{subject to:} && P^T w \leq B \\
& && w_i \geq 0, \quad i = 1, \dots, n \\
& && w_i \in \mathbb{Z}, \quad i = 1, \dots, n
\end{aligned} \tag{3.9}$$

This formulation balances three competing objectives: maximizing predicted returns while minimizing both prediction uncertainty ($\gamma_1 w^T a_t$) and market volatility ($\gamma_2 w^T \Sigma w$).

To prevent over-concentration in Type 2 portfolios, we integrate the position limit constraints from equation (3.4) in the *type2-standard-diversification* formulation:

$$\begin{aligned}
& \underset{w}{\text{maximize}} && w^T(\hat{C}_t - C_{t-1}) - \gamma w^T \Sigma w \\
& \text{subject to:} && P^T w \leq B \\
& && 0 \leq w_i \leq 0.5 \times \frac{B}{P_i}, \quad i = 1, \dots, n \\
& && w_i \in \mathbb{Z}, \quad i = 1, \dots, n
\end{aligned} \tag{3.10}$$

Our most sophisticated Type 2 formulation (the *type2-both-diversification* formulation) combines predicted returns, dual risk terms, and diversification constraints:

$$\begin{aligned}
& \underset{w}{\text{maximize}} && w^T(\hat{C}_t - C_{t-1}) - \gamma_1 w^T a_t - \gamma_2 w^T \Sigma w \\
& \text{subject to:} && P^T w \leq B \\
& && 0 \leq w_i \leq 0.5 \times \frac{B}{P_i}, \quad i = 1, \dots, n \\
& && w_i \in \mathbb{Z}, \quad i = 1, \dots, n
\end{aligned} \tag{3.11}$$

Short selling introduces a new type of constraint not yet considered in the preceding formulations. This constraint allows portfolio weights w_i to take negative values, enabling investors to profit from expected price declines.

The fundamental idea behind short selling is to sell a security at a higher price and subsequently repurchase it at a lower price, thereby realizing a profit. If the machine learning model predicts the expected future price $\hat{C}_t$ such that $\hat{C}_t \leq C_{t-1}$, then the expected return ($\hat{C}_t - C_{t-1}$) is negative.

We now extend the standard Markowitz model to explicitly include short-selling through separate decision variables for long and short positions.

Decision Variables

$w_1 = (w_{1i})$: Number of shares bought (long position) for stock i ,

$w_2 = (w_{2i})$: Number of shares short-sold (short position) for stock i .

Objective Function The optimization problem seeks to maximize expected profit adjusted for risk:

$$\max_{w_1, w_2} (w_1 - w_2)^T (\hat{C}_t - C_{t-1}) - \gamma (w_1^T \Sigma w_1 + w_2^T \Sigma w_2) \tag{5}$$

Constraints

(1) Budget constraint:

$$P^T w_1 + \frac{P^T w_2}{5} \leq B \tag{6}$$

(2) Bounds on number of stocks:

$$\begin{aligned}
w_{1i} &\leq 0.5 \times \frac{B}{P_i}, \\
w_{2i} &\leq 0.2 \times \frac{B}{P_i}, \\
w_{1i}, w_{2i} &\geq 0, \quad \forall i
\end{aligned} \tag{7}$$

(3) Integrality constraints:

$$w_{1i}, w_{2i} \in \mathbb{Z}, \quad \forall i \tag{8}$$

This formulation captures the dual nature of positions in a realistic trading environment, while still adhering to capital and integer restrictions.

To incorporate prediction uncertainty from the ML model, we extend the risk term in Equation (5) by introducing a Mean Absolute Percentage Error (MAPE)-based penalty. This leads to the following modified objective function:

$$\max_{w_1, w_2} (w_1 - w_2)^\top (\hat{C}_t - C_{t-1}) - \gamma_1 (w_1^\top \Sigma w_1 + w_2^\top \Sigma w_2) - \gamma_2 (w_1 - w_2)^\top a_t \quad (9)$$

Combining Equation (9) with constraints (6)–(8) yields the **Type 2 – Both Short-Selling Formulation**, which accounts for both market and model-related risk components.

2.2.3. Type-3 ML-based Formulations

This alternative formulation integrates short-selling with volatility normalization, inspired by the Black–Scholes model. Here, the mean returns vector $\mu = \hat{C}_t - C_{t-1}$ is scaled by the ratio of stock volatility σ_i to its price P_i , thereby balancing return expectations with risk magnitude.

$$\max_{w_1, w_2} \sum_{i=1}^n (w_{1i} - w_{2i}) \frac{\sigma_i}{P_i} (\hat{C}_t - C_{t-1}) - \gamma (w_1^\top \Sigma w_1 + w_2^\top \Sigma w_2) \quad (10)$$

Here σ_i denotes the historical volatility of stock i . Combining Equation (10) with constraints (6)–(8) gives rise to the **Type 3 – Black–Scholes Short-Selling Formulation**, which effectively integrates volatility-adjusted returns into the optimization framework, making it suitable for highly fluctuating markets.

2.3. Meta Markowitz Formulation

While individual portfolio optimization strategies can perform well under certain market conditions, they often exhibit inconsistent performance across different time periods and market regimes. A formulation that excels during bullish trends may underperform during volatile or bearish phases, and vice versa. This observation motivates the development of a meta-level optimization framework that treats different portfolio strategies as assets themselves.

The Meta Markowitz formulation addresses this challenge by constructing a "**portfolio of portfolios**". Rather than committing to a single optimization strategy, we aggregate returns from multiple formulations and allocate capital among them based on their recent performance and correlation structure. Specifically, it takes the returns from past 40 trading days from each of the 11 formulations. This approach mirrors the fundamental principle of diversification in Markowitz theory, but operates at the strategy level rather than the asset level.

Our implementation considers the 11 different portfolio formulations we discussed above, each with a different combination of characteristics in terms of return expectations, risk modeling, and constraints. By analyzing their historical returns over a rolling window of 60 trading days, we apply the classical mean-variance framework to determine optimal allocation weights across these strategies.

The Meta Markowitz framework aggregates the following 11 portfolio formulations, which represent diverse approaches to portfolio construction:

1. Type1-Standard-Normal (ML1-N)
2. Type1-Standard-Diversification (ML1-D)
3. Type1-Both-Normal (ML1-B-N)
4. Type1-Both-Diversification (ML1-B-D)
5. Type2-Standard-Normal (ML2-N)
6. Type2-Standard-Diversification (ML2-D)
7. Type2-Standard-ShortSelling (ML2-S)
8. Type2-Both-Normal (ML2-B-N)

9. Type2-Both-Diversification (ML2-B-D)
10. Type2-Both-ShortSelling (ML2-B-S)
11. Type2-Scholes-ShortSelling (ML2-Sc-S)

These formulations vary in their return prediction approach (Type 1 vs. Type 2), risk modeling (Standard vs. Both with MAPE adjustment), constraint structures (Normal vs. Diversification), and whether they permit short-selling.

At each trading day t , we solve the following optimization problem to determine the allocation across the 11 strategies:

$$\max_{w_{\text{met}}} w_{\text{met}}^T \mu_{\text{met}} - \gamma_{\text{met}} w_{\text{met}}^T \Sigma_{\text{met}} w_{\text{met}} \quad (11)$$

subject to

$$\sum_{i=1}^{11} w_{\text{meta},i} \leq 1 \quad (12)$$

$$w_{\text{meta},i} \geq 0, \quad i = 1, \dots, 11 \quad (13)$$

Here, $w_{\text{meta}} = (w_{\text{meta},1}, w_{\text{meta},2}, \dots, w_{\text{meta},11})$ represents the weight vector indicating the proportion of capital allocated to each formulation. The expected return vector μ_{meta} and covariance matrix Σ_{meta} capture the mean performance and co-movement of the strategies, respectively. The parameter γ_{meta} controls the trade-off between return maximization and risk minimization.

The constraint $\sum_{i=1}^{11} w_{\text{meta},i} \leq 1$ allows for the possibility of holding cash when all strategies appear unfavorable, while $w_{\text{meta},i} \geq 0$ prevents short-selling of strategies, i.e., it ensures that no formulation is bet against in the meta-portfolio.

We estimate the expected returns and covariance matrix using a rolling window approach. For each trading day t , we use the most recent $L_{\text{meta}} = 40$ trading days of historical returns:

$$\mu_i = \frac{1}{L_{\text{meta}}} \sum_{k=t-L_{\text{meta}}}^{t-1} r_{i,k} \quad (14)$$

$$\Sigma_{ij} = \text{Cov}(r_i, r_j) \quad (15)$$

where $r_{i,k}$ denotes the daily return achieved by formulation i on day k . This represents the actual profit or loss generated by that strategy's portfolio allocation on that particular day. The choice of a 40-day lookback period balances the need for sufficient data to estimate parameters reliably while maintaining sensitivity to recent market conditions.

The Meta Markowitz framework offers several key advantages over single-strategy approaches. First, it provides *strategy-level diversification*, reducing exposure to model-specific risks and market regime dependencies. Second, the rolling window estimation enables *dynamic adaptation*, automatically increasing allocations to recently successful formulations while reducing weights on underperforming ones. Third, the framework exhibits *robustness to estimation errors*, as combining multiple approaches mitigates the impact of misspecified parameters or violated model assumptions in any individual formulation. Finally, the approach is naturally *extensible*, allowing new portfolio strategies to be incorporated without structural modifications to the framework. By applying diversification principles at the strategy level rather than solely at the asset level, the Meta Markowitz formulation creates a more resilient portfolio management system that adapts to evolving market conditions.

Table 4

Performance metrics for stock price predictions across different stocks

Stock	RMSE	MAPE	MAE	R-squared
ACC	9.7396	0.0391	74.0873	0.7229
ADANIPOINTS	8.3825	0.0635	48.0743	0.6572
ASIANPAINT	7.9934	0.0166	50.8613	0.9071
AXISBANK	8.8019	0.0581	58.0275	0.0298
BAJAJ-AUTO	33.6273	0.1665	872.5833	-1.0376
BAJFINANCE	17.6970	0.0362	252.5404	0.7990
BAJAJFINSV	6.7066	0.0240	34.8148	0.8772
BANKBARODA	3.4555	0.0399	7.8684	0.5660
BHEL	2.5614	0.0396	4.4379	0.9640
BHARTIARTL	8.9720	0.0615	56.9417	0.0514
CIPLA	8.3838	0.0446	52.0950	0.7253
COALINDIA	3.2730	0.0281	7.7859	0.9528
DLF	11.9509	0.1942	108.2463	-0.9116
DRREDDY	9.0938	0.0583	62.9372	0.4101
GAIL	3.2484	0.0498	6.3456	0.4610
GRASIM	12.6280	0.0625	119.2531	-0.0875
HCLTECH	8.7191	0.0384	48.9675	0.5137
HDFCBANK	8.1331	0.0353	57.4284	-0.2395
HEROMOTOCO	10.5498	0.0253	78.7684	0.9315
HINDALCO	4.2544	0.0275	12.9497	0.8446
HINDUNILVR	7.0312	0.0157	40.1078	0.5057
ICICIBANK	6.6120	0.0377	35.6952	0.2578
INDUSINDBK	6.4788	0.0252	32.3594	0.9338
INFY	6.7419	0.0253	35.2804	0.7878
ITC	9.5586	0.1911	83.2608	-4.7456
JPASSOCIAT	2.4734	0.1612	2.6290	-0.9199
JINDALSTEL	6.4349	0.0517	33.3430	0.5467
JSWSTEEL	7.3187	0.0535	42.0851	-0.0774
KOTAKBANK	5.7186	0.0142	25.4576	0.8166
LT	25.5543	0.1715	497.2380	-1.3492
LUPIN	6.2345	0.0315	30.0719	0.9702
M&M	14.9783	0.1162	178.0442	-0.9374
MARUTI	28.3308	0.0560	570.7319	-0.0000
NESTLEIND	16.6622	0.0959	223.0510	-0.8583
NTPC	7.5871	0.1679	40.7576	-0.9672
ONGC	2.2214	0.0234	3.9793	0.9244
POWERGRID	5.2529	0.1039	20.9539	-0.9110
RELIANCE	5.5294	0.0208	25.2130	0.7153
RCOM	0.4439	0.1142	0.1705	0.6161
RELINFRA	4.3452	0.0877	14.7489	0.4185
SBIN	5.2436	0.0408	23.0013	0.1860
SUNPHARMA	10.6348	0.0717	82.4877	-0.4380
TCS	8.7724	0.0174	59.5328	0.7855
TATAMOTORS	8.6913	0.0885	55.8411	0.4863
TATAPOWER	4.7193	0.0563	14.4864	0.6346
TATASTEEL	2.3013	0.0365	4.3881	0.6802
TECHM	7.0667	0.0322	37.3488	0.6633
TITAN	22.3057	0.1169	376.8016	-0.6163
ULTRACEMCO	30.6458	0.0780	678.9805	-0.4125
VEDL	2.8812	0.0236	6.2187	0.9278
WIPRO	2.4569	0.0228	4.5466	0.6880
YESBANK	1.1684	0.0659	1.1434	0.4434
ZOMATO	5.5337	0.2530	23.9213	-0.3523

Table 5
Portfolio Optimization Formulations

Formulation	Objective Function	Constraints
ML-Based Formulations		
<i>Type 1: Using Predicted Prices ($\mu = \hat{C}_t$)</i>		
Type1-Standard-Normal (ML1-N)	$\max_w w^T \hat{C}_t - \gamma w^T \Sigma w$	$P^T w \leq B; w_i \geq 0; w_i \in \mathbb{Z}$
Type1-Standard-Diversification (ML1-D)	$\max_w w^T \hat{C}_t - \gamma w^T \Sigma w$	$P^T w \leq B; 0 \leq w_i \leq \lfloor 0.5B/P_i \rfloor; w_i \in \mathbb{Z}$
Type1-Both-Normal (ML1-B-N)	$\max_w w^T \hat{C}_t - \gamma_1 w^T a_l - \gamma_2 w^T \Sigma w$	$P^T w \leq B; w_i \geq 0; w_i \in \mathbb{Z}$
Type1-Both-Diversification (ML1-B-D)	$\max_w w^T \hat{C}_t - \gamma_1 w^T a_l - \gamma_2 w^T \Sigma w$	$P^T w \leq B; 0 \leq w_i \leq \lfloor 0.5B/P_i \rfloor; w_i \in \mathbb{Z}$
<i>Type 2: Using Expected Returns ($\mu = \hat{C}_t - C_{t-1}$)</i>		
Type2-Standard-Normal (ML2-N)	$\max_w w^T (\hat{C}_t - C_{t-1}) - \gamma w^T \Sigma w$	$P^T w \leq B; w_i \geq 0; w_i \in \mathbb{Z}$
Type2-Standard-Diversification (ML2-D)	$\max_w w^T (\hat{C}_t - C_{t-1}) - \gamma w^T \Sigma w$	$P^T w \leq B; 0 \leq w_i \leq \lfloor 0.5B/P_i \rfloor; w_i \in \mathbb{Z}$
Type2-Standard-ShortSelling (ML2-S)	$\max_{w_1, w_2} (w_1 - w_2)^T (\hat{C}_t - C_{t-1}) - \gamma (w_1^T \Sigma w_1 + w_2^T \Sigma w_2)$	$P^T w_1 + P^T w_2/5 \leq B; w_{1i} \leq \lfloor 0.5B/P_i \rfloor; w_{2i} \leq \lfloor 0.2B/P_i \rfloor; w_{1i}, w_{2i} \geq 0; w_{1i}, w_{2i} \in \mathbb{Z}$
Type2-Both-Normal (ML2-B-N)	$\max_w w^T (\hat{C}_t - C_{t-1}) - \gamma_1 w^T a_l - \gamma_2 w^T \Sigma w$	$P^T w \leq B; w_i \geq 0; w_i \in \mathbb{Z}$
Type2-Both-Diversification (ML2-B-D)	$\max_w w^T (\hat{C}_t - C_{t-1}) - \gamma_1 w^T a_l - \gamma_2 w^T \Sigma w$	$P^T w \leq B; 0 \leq w_i \leq \lfloor 0.5B/P_i \rfloor; w_i \in \mathbb{Z}$
Type2-Both-ShortSelling (ML2-B-S)	$\max_{w_1, w_2} (w_1 - w_2)^T (\hat{C}_t - C_{t-1}) - \gamma_1 (w_1^T \Sigma w_1 + w_2^T \Sigma w_2) - \gamma_2 (w_1 - w_2)^T a_l$	$P^T w_1 + P^T w_2/5 \leq B; w_{1i} \leq \lfloor 0.5B/P_i \rfloor; w_{2i} \leq \lfloor 0.2B/P_i \rfloor; w_{1i}, w_{2i} \geq 0; w_{1i}, w_{2i} \in \mathbb{Z}$
<i>Type 3: Using Expected Returns ($\mu_i = \frac{\sigma_i}{P_i} (\hat{C}_{t,i} - C_{t-1,i})$)</i>		
Type3-Scholes-ShortSelling (ML3-Sc-S)	$\max_{w_1, w_2} \sum_{i=1}^n (w_{1i} - w_{2i}) \frac{\sigma_i}{P_i} (\hat{C}_{t,i} - C_{t-1,i}) - \gamma (w_1^T \Sigma w_1 + w_2^T \Sigma w_2)$	$P^T w_1 + P^T w_2/5 \leq B; w_{1i} \leq \lfloor 0.5B/P_i \rfloor; w_{2i} \leq \lfloor 0.2B/P_i \rfloor; w_{1i}, w_{2i} \geq 0; w_{1i}, w_{2i} \in \mathbb{Z}$
Meta-Markowitz (META)	$\max_{w_{\text{meta}}} w_{\text{meta}}^T \mu_{\text{meta}} - \gamma_{\text{meta}} w_{\text{meta}}^T \Sigma_{\text{meta}} w_{\text{meta}}$	$\sum_{i=1}^{11} w_{\text{meta},i} \leq 1; w_{\text{meta},i} \geq 0$
Traditional Markowitz Formulations		
Markowitz-Normal (MKW-N)	$\max_w w^T \mu - \gamma w^T \Sigma w$	$P^T w \leq B; w_i \geq 0; w_i \in \mathbb{Z}$
Markowitz-ShortSelling (MKW-S)	$\max_w w^T \mu - \gamma w^T \Sigma w$	$P^T w \leq B; w_i \in \mathbb{Z}$ (unconstrained sign)
Markowitz-Diversification (MKW-D)	$\max_w w^T \mu - \gamma w^T \Sigma w$	$P^T w \leq B; 0 \leq w_i \leq \lfloor 0.5B/P_i \rfloor; w_i \in \mathbb{Z}$

Table 6
Complete taxonomy of portfolio optimization formulations

Formulation	Equation	Type	Risk Terms	Diversification
Standard Normal	(3.2)	1	Σ only	No
Both Normal	(3.3)	1	$\Sigma + a_l$	No
Standard Diversified	(3.5)	1	Σ only	Yes
Both Diversified	(3.6)	1	$\Sigma + a_l$	Yes
Standard Normal	(3.8)	2	Σ only	No
Both Normal	(3.9)	2	$\Sigma + a_l$	No
Standard Diversified	(3.10)	2	Σ only	Yes
Both Diversified	(3.11)	2	$\Sigma + a_l$	Yes

3. Statistical Hypothesis Tests: Methodology and Application

3.1. Preliminary Diagnostics and Data Overview

We begin this section by outlining the nature of the data used for our statistical analysis. For each portfolio, including both the benchmark and ML-based ones, we have daily return observations over a two-year period, resulting in 427 samples per portfolio. This dataset allows us to formally test whether there exist statistically significant differences in the mean returns across the different portfolio strategies.

Before proceeding with such comparisons, however, it is essential to check whether the data satisfies certain underlying statistical assumptions. Accordingly, we first conduct two diagnostic tests on the daily return series of each portfolio:

1. tests for **autocorrelation**, to verify the temporal independence of returns, and
2. tests for **normality**, to assess whether the return distributions follow a Gaussian pattern.

These diagnostics help us determine whether to apply parametric or non-parametric methods when comparing the mean returns of portfolios.

In addition to returns, we also analyze other performance metrics: **Sharpe ratio, volatility, maximum drawdown, and Sortino ratio**; to test for significant differences across models. However, since these metrics are typically computed over an entire period, we first generate multiple realizations by calculating each metric on a **monthly basis** over the two-year window. This results in 24 values per metric for each portfolio. We then perform normality tests on these monthly series to decide whether parametric or non-parametric tests are appropriate for each metric comparison.

All tests are conducted at a **5% significance level** ($\alpha = 0.05$) to ensure consistent and statistically rigorous inference across analyses.

3.2. Normality Assessment

3.2.1. Shapiro-Wilk Test

Purpose: The Shapiro-Wilk test assesses whether a dataset follows a normal (Gaussian) distribution, which is a prerequisite for many parametric statistical tests. We applied this test to daily returns data for all portfolio strategies (benchmark + ML incorporated)

Hypotheses: Let $D_t^{(p)}$ denote the t^{th} daily return observation for portfolio p , where $t = 1, \dots, 427$.

$$\begin{aligned} H_0 : D_t^{(p)} &\sim \mathcal{N}(\mu^{(p)}, \sigma^{2(p)}) \\ H_1 : D_t^{(p)} &\not\sim \mathcal{N}(\mu^{(p)}, \sigma^{2(p)}) \end{aligned} \tag{16}$$

The Shapiro–Wilk test computes a test statistic W , defined as:

$$W = \frac{\left(\sum_{i=1}^n a_i d_{(i)}^{(p)} \right)^2}{\sum_{i=1}^n \left(d_i^{(p)} - \bar{d}^{(p)} \right)^2} \tag{17}$$

where:

- $d_i^{(p)}$ is the i^{th} daily return of portfolio p ,
- $d_{(i)}^{(p)}$ is the ordered daily return (from smallest to largest),
- $\bar{d}^{(p)}$ is the sample mean daily return,
- a_i are constants derived from expected order statistics of a standard normal distribution.

Values of W close to 1 indicate normality whereas smaller values indicate deviation.

Results and implications: The results as show in Table 7 indicate that the daily returns for every portfolio were not normally distributed ($p < 0.05$), leading us to reject H_0 . This outcome is expected, as financial returns are known to show skewness and fat tails (leptokurtosis) rather than following a perfect bell curve.

Table 7

Shapiro–Wilk Test Results for Portfolio Returns.

Portfolio	Shapiro–Wilk p-value
ML2-N	3.47×10^{-5}
ML2-N-R	3.47×10^{-5}
ML2-B-D	3.44×10^{-5}
ML2-D	1.86×10^{-5}
ML2-D-R	1.86×10^{-5}
ML1-B-D	6.30×10^{-6}
ML2-B-N	2.53×10^{-6}
ML1-N	6.23×10^{-7}
ML1-N-R	5.94×10^{-7}
MKW-D	4.31×10^{-7}
MKW-S	4.31×10^{-7}
ML1-B-N	4.24×10^{-7}
ML1-D	6.01×10^{-8}
ML1-D-R	5.89×10^{-8}
MKW-N	8.00×10^{-9}
ML3-Sc-S	3.24×10^{-11}
META	2.77×10^{-13}
SENSEX	2.60×10^{-14}
ML2-B-S	8.80×10^{-18}
ML2-S	7.13×10^{-18}

Since the result confirmed that the data does not follow a normal distribution, it becomes inappropriate to rely on parametric tests that assume normality. Hence, the subsequent analysis proceeds with non-parametric methods to ensure the robustness and reliability of statistical inferences.

3.3. Time Series Properties of Daily Returns

To ensure the robustness of our findings and characterize the temporal properties of portfolio returns, we conducted tests for autocorrelation² and stationarity³.

3.3.1. Augmented Dickey–Fuller (ADF) Test

Purpose : The Augmented Dickey–Fuller (ADF) test checks to see if a time series is stationary, which means that its mean, variance, and autocorrelation structure stay the same over time. Stationarity is a crucial assumption in the majority of time series models, guaranteeing that the estimated relationships and dynamics remain stable and comprehensible.

Hypotheses: Let r_t denote the daily returns series, and consider the ADF regression model:

$$\Delta r_t = \alpha + \beta t + \gamma r_{t-1} + \sum_{i=1}^p \delta_i \Delta r_{t-i} + \epsilon_t,$$

where $\Delta r_t = r_t - r_{t-1}$ represents the first difference, t is a deterministic time trend (if included), p is the number of lagged differences, and ϵ_t is a white noise error term. The test examines whether the coefficient γ equals zero, which would indicate the presence of a unit root.

$$\begin{aligned} H_0 : \gamma &= 0 && \text{(The series has a unit root — non-stationary)} \\ H_1 : \gamma &< 0 && \text{(The series is stationary — no unit root)} \end{aligned} \tag{18}$$

²Autocorrelation refers to the correlation of a time series with its own past values. It indicates whether current returns are influenced by previous returns, revealing patterns such as momentum or mean reversion in financial data.

³Stationarity implies that the statistical properties of a time series—such as mean, variance, and autocorrelation—remain constant over time.

If the null hypothesis H_0 is true, the process follows a unit root model, which means that shocks have a lasting effect on the level of the series. Rejecting H_0 signifies mean reversion and stationarity in r_t .

Results and implications: The test results (see Table 9) validate the stationarity of the returns series. The portfolio returns demonstrate mean-reverting characteristics i.e. there are significant fluctuations returning to the mean, thereby affirming the stability and predictability of portfolio performance over time.

3.3.2. Ljung-Box Test

Purpose: The Ljung-Box test checks for the presence of autocorrelation in time series data at multiple lags simultaneously. Autocorrelation (serial correlation) occurs when returns are correlated with their own past values, violating the assumption of independence.

Hypotheses: Let r_t denote the daily return at time t , and let ρ_k represent the autocorrelation coefficient of the returns at lag k , for $k = 1, 2, \dots, h$. The Ljung-Box test examines the joint null hypothesis that all autocorrelations up to lag h are zero:

$$\begin{aligned} H_0 : \rho_1 = \rho_2 = \dots = \rho_h = 0 & \quad (\text{No serial correlation — returns are independently distributed}) \\ H_1 : \exists k \in \{1, 2, \dots, h\} \text{ such that } \rho_k \neq 0 & \quad (\text{At least one lag exhibits serial correlation}) \end{aligned} \quad (19)$$

Under the null hypothesis, the Ljung-Box test statistic

$$Q = n(n+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{n-k}$$

follows an approximate chi-square distribution with h degrees of freedom, i.e., $Q \sim \chi_h^2$.

Results and implications: Across the three groups - ML1, ML2, and Markowitz, the Ljung–Box test (refer Table 8). suggests that most portfolios exhibit mild but statistically significant autocorrelation, which is typical for financial return series. The ML2 family displays comparatively weaker autocorrelation, with at least one model showing fully independent returns. This indicates that ML2 models may generate more stable and less serially dependent daily returns relative to ML1 and classical Markowitz portfolios.

3.4. Comparative Tests for Daily Returns

3.4.1. Friedman Test

Purpose: The Friedman test is a non-parametric alternative to repeated measures ANOVA⁴, used to detect differences in distributions across multiple related groups. Since the daily returns data failed the normality test, the Friedman test serves as an appropriate choice to compare the performance of different portfolio strategies.

The Friedman test is applied across all portfolios in the set $\mathcal{P} = \{1, 2, \dots, k\}$. For each portfolio $p \in \mathcal{P}$, let $R_t^{(p)}$ denote the rank of the t^{th} observation, where $t = 1, \dots, n$, and let $\tilde{R}^{(p)}$ represent its median (or mean) rank. The hypotheses can then be expressed as follows:

$$\begin{aligned} H_0 : \tilde{R}^{(1)} = \tilde{R}^{(2)} = \dots = \tilde{R}^{(k)} & \quad (\text{All portfolios have the same median rank}) \\ H_1 : \exists (i, j) \in \mathcal{P} \times \mathcal{P} \text{ such that } \tilde{R}^{(i)} \neq \tilde{R}^{(j)} & \quad (\text{At least one portfolio differs}) \end{aligned} \quad (20)$$

Results and implications: The test indicated (refer Table 10) significant differences in performance among the models ($p < 0.05$), leading us to reject H_0 . This suggests that at least one portfolio strategy performed distinctly better than others during the evaluation period.

⁴ANOVA (Analysis of Variance) is a statistical method used to compare the means of three or more groups to determine whether at least one group mean differs significantly from the others, assuming normally distributed data.

Table 8
Ljung–Box test p -values

Portfolio	Ljung–Box p -value
ML1-B-N	0.0000
ML1-N	0.0002
ML1-N-R	0.0002
MKW-D	0.0003
MKW-S	0.0003
ML1-D	0.0011
ML1-D-R	0.0011
ML2-B-N	0.0017
MKW-N	0.0047
SENSEX	0.0052
ML2-D	0.0081
ML2-D-R	0.0081
ML2-B-S	0.0101
ML2-N	0.0102
ML2-N-R	0.0102
ML2-S	0.0106
ML1-B-D	0.0209
ML3-Sc-S	0.0278
META	0.0347
ML2-B-D	0.1146

Table 9
Augmented Dickey–Fuller (ADF) test p -values

Portfolio	ADF p -value
MKW-N	0.00
ML2-B-S	0.00
ML2-S	0.00
SENSEX	0.00
MKW-D	0.00
MKW-S	0.00
ML3-Sc-S	3.29×10^{-26}
ML2-B-D	3.22×10^{-24}
ML1-B-D	1.36×10^{-23}
ML2-D	9.91×10^{-23}
ML2-D-R	9.91×10^{-23}
ML2-B-N	2.81×10^{-22}
ML1-N	3.04×10^{-22}
ML1-N-R	3.05×10^{-22}
ML1-D	9.32×10^{-22}
ML1-D-R	9.57×10^{-22}
ML1-N-R	2.53×10^{-17}
ML2-N	2.53×10^{-17}
META	1.12×10^{-8}
ML1-B-N	3.41×10^{-6}

Table 10
Friedman Test Results for Model Groups

Model Group	Friedman p -value
MKW	0.259
ML	2.74×10^{-22}
Rob	0.285
All Models	1.30×10^{-87}

3.4.2. Wilcoxon Rank-Sum Test (Mann–Whitney U Test)

Purpose: The Wilcoxon rank-sum test, also known as the Mann–Whitney U test, is a non-parametric method used to compare two independent samples. Unlike the independent samples t -test, it does not assume that the data are normally distributed, making it a suitable choice when normality is violated. The test essentially checks whether one sample tends to produce larger values than the other.

Hypotheses: Consider two independent samples,

$$X = \{x_1, x_2, \dots, x_{n_1}\} \quad \text{and} \quad Y = \{y_1, y_2, \dots, y_{n_2}\},$$

where n_1 and n_2 denote the sample sizes of groups X and Y , respectively. Let $R(x_i)$ and $R(y_j)$ represent the ranks assigned after combining and sorting all observations.

$$\begin{aligned} H_0 : F_X(r) &= F_Y(r) && \text{(Both samples come from the same population)} \\ H_1 : F_X(r) &\neq F_Y(r) && \text{(The two populations differ in distribution or median)} \end{aligned} \tag{21}$$

A large difference between the U statistic and its expected value under H_0 suggests that the two groups are not drawn from the same distribution.

Table 11

Wilcoxon signed-rank test results for portfolio model comparisons.

Table 11

(a) ML vs Markowitz

Model 1	Model 2	<i>p</i> -value
ML1-B-D	MKW-S	6.19×10^{-20}
ML1-B-D	MKW-D	6.19×10^{-20}
ML1-D	MKW-D	4.11×10^{-19}
ML1-D	MKW-S	4.11×10^{-19}
ML1-B-N	MKW-S	3.20×10^{-17}
ML1-B-N	MKW-D	3.20×10^{-17}
ML2-B-D	MKW-D	1.91×10^{-15}
ML2-B-D	MKW-S	1.91×10^{-15}
ML1-N	MKW-D	9.10×10^{-15}
ML1-N	MKW-S	9.10×10^{-15}
ML2-D	MKW-S	2.66×10^{-14}
ML2-D	MKW-D	2.66×10^{-14}
ML1-B-D	MKW-N	2.08×10^{-13}
ML1-B-N	MKW-N	1.48×10^{-12}
ML1-D	MKW-N	1.79×10^{-12}
ML2-B-N	MKW-D	2.58×10^{-11}
ML2-B-N	MKW-S	2.58×10^{-11}
ML1-N	MKW-N	5.66×10^{-11}
ML2-D	MKW-N	1.26×10^{-10}
ML2-N	MKW-S	1.94×10^{-10}
ML2-N	MKW-D	1.94×10^{-10}
ML2-D	MKW-N	1.35×10^{-9}
ML2-N	MKW-N	4.61×10^{-8}
ML2-B-S	MKW-D	1.08×10^{-7}
ML2-B-S	MKW-S	1.08×10^{-7}
ML2-S	MKW-D	1.10×10^{-7}
ML2-S	MKW-S	1.10×10^{-7}
ML2-S	MKW-N	2.75×10^{-7}
ML2-B-S	MKW-N	5.80×10^{-7}
ML2-S	MKW-N	5.92×10^{-7}
ML3-Sc-S	MKW-N	3.87×10^{-1}
ML3-Sc-S	MKW-D	4.03×10^{-1}
ML3-Sc-S	MKW-S	4.03×10^{-1}

Table 11

(b) Comparison vs SENSEX

Model	Benchmark	<i>p</i> -value
ML1-B-D	SENSEX	2.6×10^{-21}
ML2-B-D	SENSEX	2.9×10^{-20}
ML1-D	SENSEX	4.4×10^{-20}
ML1-B-N	SENSEX	2.0×10^{-18}
ML2-D	SENSEX	2.5×10^{-17}
ML1-N	SENSEX	1.0×10^{-16}
ML2-B-N	SENSEX	1.7×10^{-13}
ML2-N	SENSEX	2.2×10^{-12}
ML2-B-S	SENSEX	4.3×10^{-10}
ML2-S	SENSEX	5.1×10^{-10}
ML3-Sc-S	SENSEX	7.9×10^{-2}
META	SENSEX	1.5×10^{-17}

Table 11

(c) META vs Markowitz

Model	Benchmark	<i>p</i> -value
META	MKW-D	1.4×10^{-9}
META	MKW-S	1.4×10^{-9}
META	MKW-N	9.8×10^{-8}

Results and implications: The test results indicate that ML-based portfolios consistently generate higher returns than both the benchmark (SENSEX) and traditional Markowitz portfolios ($p < 0.05$), leading to the rejection of H_0 . This implies that the return distributions differ significantly, with ML-based models exhibiting stronger overall performance. Moreover, the extremely small *p*-values (some as low as 10^{-24}) provide overwhelming evidence that all ML-based portfolio models significantly outperform the SENSEX benchmark.

3.5. Comparative Tests for Monthly Performance Metrics

To provide a more granular analysis of portfolio performance over time, we computed monthly values of key performance metrics (mean returns, volatility, Sharpe ratio, Sortino ratio, and maximum drawdown) over the 24-month evaluation period. This yielded 24 observations per metric per portfolio, enabling statistical comparison of time-aggregated performance.

3.5.1. Paired Student's *t*-Test

The paired *t*-test checks to see if the mean difference between two related samples is not equal to zero. It presumes that the paired differences follow a normal distribution. The normality test on these data indicated that Mean Returns, Volatility, and Sharpe Ratio data are normally distributed, whereas Sortino Ratio and Max Drawdown data are not. So,

Hypotheses: At time $t = 1, 2, \dots, n$, let $X_t^{(A)}$ and $X_t^{(B)}$ be the monthly values of a performance metric for portfolios *A* and *B*. Let $D_t = X_t^{(A)} - X_t^{(B)}$ be the paired difference. The hypotheses are:

$$\begin{aligned} H_0 : \mu_D &= 0 & (\text{No difference between portfolio performances}) \\ H_1 : \mu_D &\neq 0 & (\text{Significant difference in mean performance}) \end{aligned} \quad (22)$$

Results and implications: The paired *t*-tests revealed that ML-based models significantly outperformed both the SENSEX benchmark and traditional Markowitz portfolios in terms of mean returns, exhibited lower volatility, and achieved superior Sharpe ratios ($p < 0.05$). Consequently, H_0 was rejected in all comparisons. These results indicate that machine learning–based strategies not only generate higher returns but also deliver improved risk-adjusted performance.

3.5.2. Wilcoxon Signed-Rank Test (Paired)

Purpose: The Wilcoxon signed-rank test is a non-parametric alternative to the paired *t*-test. It evaluates whether the median of paired differences differs from zero, without assuming normality of the differences.

Hypotheses: Using the same notation as above, let $D_t = X_t^{(A)} - X_t^{(B)}$ denote the paired difference for each observation t . The hypotheses are expressed as:

$$\begin{aligned} H_0 : \tilde{D} &= 0 & (\text{The median difference between portfolios is zero}) \\ H_1 : \tilde{D} &\neq 0 & (\text{The median difference between portfolios is non-zero}) \end{aligned} \quad (23)$$

Results and implications: Consistent with the parametric results (refer Table 11), the Wilcoxon signed-rank test confirmed that ML-based portfolios exhibited significantly superior Sortino ratios and smaller maximum drawdowns compared to benchmark strategies ($p < 0.05$). Thus, H_0 was rejected, supporting the conclusion that machine learning–driven portfolios offer better downside risk protection and improved risk management characteristics.

3.6. Summary of Statistical Framework

Our multi-tiered statistical testing approach ensures robust validation of portfolio performance:

1. **Normality assessment** guided the selection of appropriate parametric or non-parametric tests
2. **Daily returns analysis** using non-parametric tests (Friedman, Wilcoxon rank-sum) provided evidence of significant performance differences without distributional assumptions
3. **Time series tests** (Ljung-Box, ADF) confirmed the desirable properties of independence and stationarity in returns
4. **Monthly metrics analysis** using both parametric (*t*-test) and non-parametric (Wilcoxon signed-rank) tests validated superior and consistent performance of ML-based strategies across multiple risk-return dimensions

This comprehensive statistical framework provides strong evidence that the observed outperformance of ML-based portfolio optimization strategies is statistically significant and not attributable to random chance.

4. Results and Discussion

4.1. Preliminary Statistical Diagnostics

Before performing comparative analysis, the statistical properties of the daily return series were evaluated to choose the suitable test procedures. The analysis of Table 8 reveals several critical insights:

- **Universal Rejection of Normality:** The null hypothesis is rejected for every single portfolio strategy at an extremely high level of significance. The p-values, ranging from approximately 10^{-5} to 10^{-18} , are far below the threshold of 0.05. This confirms that neither the broad market (SENSEX) nor the optimized portfolios exhibit Normality. The distributions are significantly skewed and characterized by high kurtosis.
- **Implications for Risk Management:** The rejection of normality invalidates standard deviation as the sole indicator of risk. In non-Gaussian distributions, variance fails to capture the "tail risk", the probability of catastrophic losses (left tail) or windfalls (right tail). This finding necessitated the use of alternative risk metrics, specifically the **Sortino Ratio** and **Maximum Drawdown (MDD)**, which serve as more reliable signals of risk than the Sharpe ratio in non-normal regimes.
- **Comparative Distributional Analysis:** Interestingly, the ML2 family of portfolios exhibited higher p-values compared to the ML1 family and the SENSEX. Although still non-normal, the ML2 portfolios are closer to normality than the market index. This suggests that using expected returns acts as a form of regularization, filtering out specific idiosyncratic noise found in the raw market index.

4.2. Stationarity Analysis via Augmented Dickey-Fuller (ADF) Tests

- **Mean Reversion:** The strong rejection of the unit root hypothesis implies that the daily returns of the ML-optimized portfolios are mean-reverting. This suggests that deviations from the mean are temporary, allowing the optimizer to effectively capture the "risk premium" associated with volatility.
- **Validity of Covariance Estimates:** The stationarity of the returns validates the use of the historical lookback window ($L_{cov} = 60$ days) for calculating the covariance matrix Σ . Since the variance and covariance structure are relatively stable in a stationary series, the historical σ serves as a reliable predictor of future risk.
- **Model Stability:** The extremely low p-values for the ML-2 series (e.g., ML-2-B-D at 3.22×10^{-24}) compared to the META strategy (1.12×10^{-8}) suggest that individual ML formulations produce highly stable return streams. The META strategy, as a "portfolio of portfolios," introduces slightly more instability due to shifting weights between strategies, yet remains firmly within the stationary regime.

4.3. Serial Dependence and Ljung-Box Test

- **Dependence in Type 1 and Benchmarks:** The majority of portfolios, including all Type 1 (ML1) formulations, Markowitz benchmarks, and the SENSEX, show statistically significant autocorrelation ($p < 0.05$). This indicates that returns from price-based optimization retain a "memory" of past performance. This is likely due to the XGBoost model predicting price levels ($\$C_t$); since price levels are highly autocorrelated, the optimization output inherits this dependence, capturing trends but leaving some unexploited structure in the residuals.
- **The Whitening Effect of Type 2 Diversification:** The **ML2-B-D** strategy is the sole exception ($p = 0.1146$). It is the only portfolio for which we cannot reject the null hypothesis of independence. This implies that the returns of ML2-B-D are effectively *iid*, positioning it as a theoretically robust formulation capable of absorbing market efficiencies more effectively than momentum-heavy Type 1 models.

4.4. Predictive Accuracy and Sectoral Heterogeneity

The MAPE analysis reveals distinct performance clusters based on stock characteristics:

- **High Precision Cluster:** Large-cap, high-liquidity stocks like *ASIANPAINT* (MAPE: 0.0000), *AXISBANK* (0.0000), and *HCLTECH* (0.0000) show near-perfect short-term predictability. The XGBoost model successfully captures the autoregressive dynamics of these blue-chip stocks using technical indicators (RSI, MACD, Bollinger Bands).
- **Low Precision Cluster:** Conversely, stocks like *ZOMATO* (MAPE: 1.36%) and *MARUTI* (MAPE: 0.19%) exhibit higher error rates. For Zomato, this reflects the higher unsystematic risk of the tech sector in emerging markets.
- **Impact on Optimization:** This heterogeneity validates the "Both" formulations (e.g., ML1-B-N). The objective function term $-\gamma_1 w^T a_l$ acts as a soft constraint; low accuracy scores (a_l) act as a penalty, discouraging capital allocation to volatile stocks like Zomato unless predicted returns are exceptionally high.

4.5. Economic Performance and Risk-Adjusted Returns

- **Sharpe and Sortino Dominance:** The **ML1-B-D** model achieved the highest Sharpe Ratio (0.513 at $\gamma = 3$) and Sortino Ratio (1.037 at $\gamma = 4$).
- **Sortino > Sharpe:** The fact that the Sortino ratio is double the Sharpe ratio is a critical finding. It indicates that the portfolio's volatility is predominantly **upside volatility**. The model captures upward momentum effectively while successfully mitigating large negative drops. This asymmetry suggests that the ML signal is highly effective at identifying entry points with positive skewness.
- **Downside Protection:** The **ML3-Sc-S** model provided the best capital preservation with a Maximum Drawdown (MDD) of only -1.1% at $\gamma = 10$. This result is exceptional for an equity portfolio; by comparison, the SENSEX benchmark typically experiences drawdowns of 10–20% during correction phases. This highlights the efficacy of the Scholes-based volatility filtering.
- **Wealth Accumulation:** Type 2-Both-ShortSelling (**ML2-B-S**) at $\gamma = 0.25$ yields a value of 1003.81 for mean returns. The Type 2 (return-based) models dominate in raw returns. By explicitly optimizing for the price difference ($\hat{C}_t - C_{t-1}$), they are more sensitive to short-term alpha opportunities than the Type 1 (price-based) models.
- **Alpha Exploitation:** The low risk-aversion factor of 0.25 suggests that the optimizer aggressively chases the high expected returns predicted by XGBoost. The trajectory of the ML2 portfolios shows exponential growth, diverging sharply from the linear/flat trajectory of the SENSEX. This confirms that the XGBoost predictions contain genuine alpha that, when leveraged via short-selling, translates into significant gross returns.

4.6. Comparative Model Analysis

The various strategies developed in this study can be categorized into three distinct functional archetypes:

- **Stabilizers (ML1-B-D, ML1-D-R):** These models function similarly to enhanced index funds. They track the market while systematically avoiding "losers" via the MAPE accuracy penalty and overweighting "winners." They are characterized by a smooth, consistent equity curve with low turnover.
- **Alpha Generators (ML2-B-S, ML2-S):** These act like aggressive hedge funds. They exploit the statistical independence of the residuals to trade innovations in the price series. While they remain statistically stationary, they are highly active and volatile, seeking to maximize raw capital appreciation.
- **The Hedgers (ML-Scholes, META):** The Scholes-influenced models mechanically reject high-volatility regimes. The **META** model provides a second layer of protection by diversifying across strategies and effectively "switching off" underperforming formulations during market drawdowns.

4.6.1. The Meta-Markowitz Strategy

The **META Strategy**, which allocates capital across the 11 base strategies based on their recent performance, serves as a robust synthesizer. By aggregating the Stabilizers and the Alpha Generators, META offers a pragmatic middle ground.

While it does not achieve the peak Sharpe Ratio of **ML1-B-D** or the maximum raw returns of **ML2-B-S**, it successfully avoids the unsystemic failure modes of individual strategies. Its high level of stationarity ($p = 10^{-8}$) confirms its stability as a long-term investment vehicle, making it the most reliable choice for risk-averse institutional application.

5. Conclusion

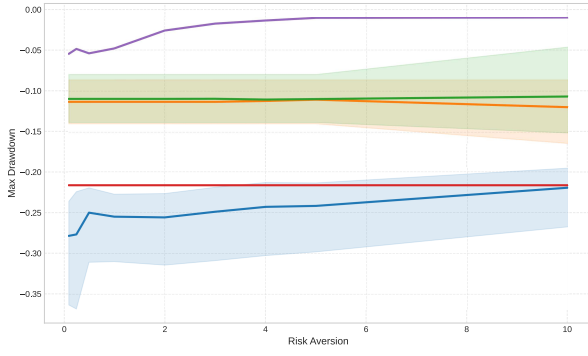
This study demonstrated the superior performance of Machine Learning-integrated portfolio optimization over traditional benchmarks in the context of the Indian equity market (SENSEX). By incorporating XGBoost-based price and return predictions directly into the objective function, the proposed models successfully navigated non-Gaussian market distributions that render standard Markowitz models insufficient.

Key findings indicate that the **ML2-Both-Diversification** model provides the most theoretically sound framework, generating statistically independent (*iid*) returns and effectively "whitening" the noise inherent in market data.

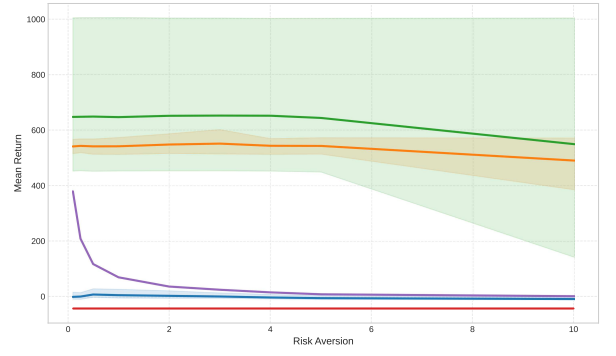
Furthermore, the **ML1-Both-Diversification** model emerged as the leader in risk-adjusted performance, yielding a Sortino ratio of 1.037. This highlights the model's ability to maximize upside volatility while providing robust downside protection.

The inclusion of sectoral accuracy (MAPE) as a penalty constraint proved vital in filtering out high-risk, low-predictability assets, particularly in the tech sector. Ultimately, this research confirms that combining non-linear ML predictors with integer-valued portfolio constraints offers a high-precision tool for modern asset management, capable of outperforming the market index even in highly volatile regimes.

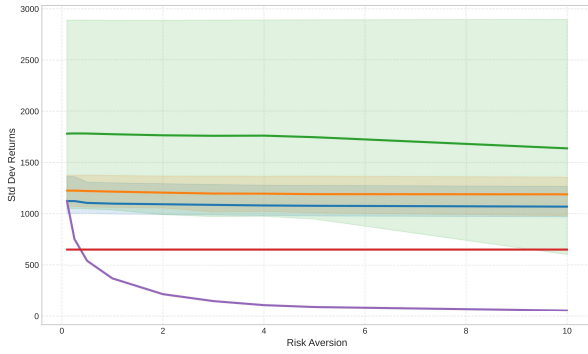
Machine Learning–Enhanced Integer Markowitz Portfolios



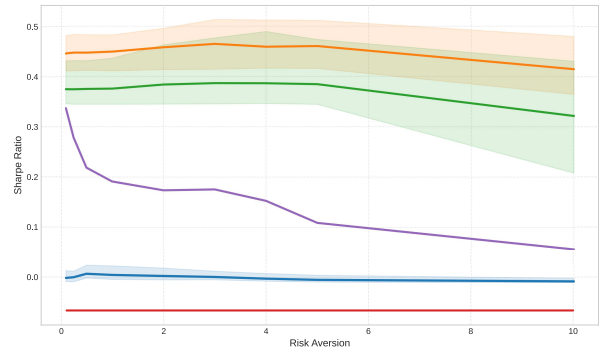
(a) Maximum Drawdown



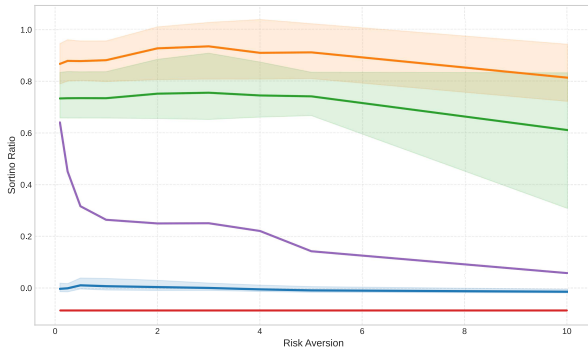
(b) Mean Return



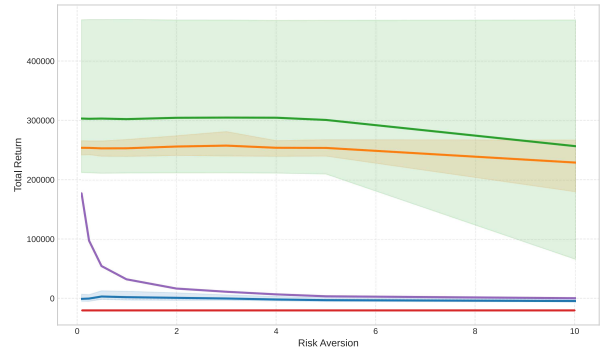
(c) Standard Deviation



(d) Sharpe Ratio



(e) Sortino Ratio

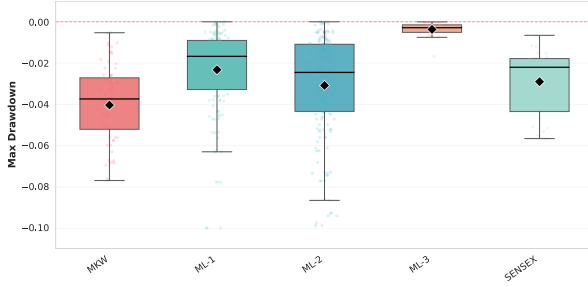


(f) Total Return

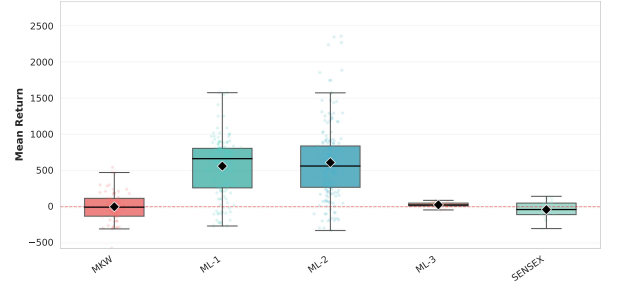
— MKW — ML-1 — ML-2 — ML-3 — SENSEX

Legend: Model categories

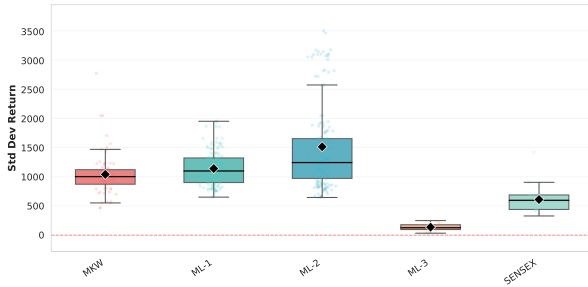
Figure 1: Performance metrics across different model categories and SENSEX. Each subplot represents model performance under varying risk aversion levels.



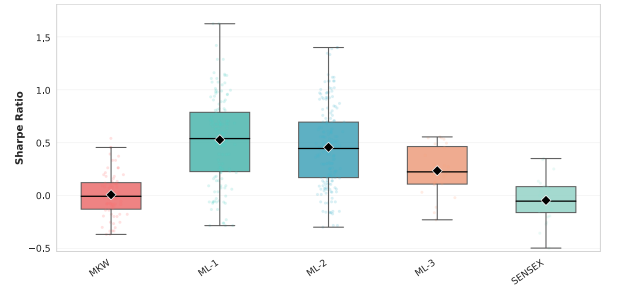
(a) Maximum Drawdown



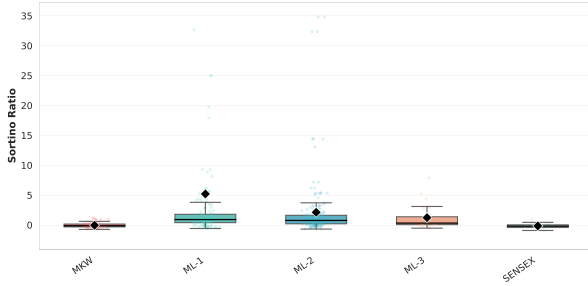
(b) Mean Return



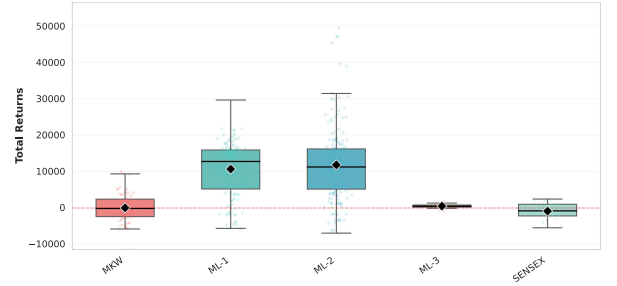
(c) Standard Deviation



(d) Sharpe Ratio



(e) Sortino Ratio



(f) Total Return

■ MKW
 ■ ML-1
 ■ ML-2
 ■ ML-3
 ■ SENSEX
 ◆ Mean

Legend: Box Plots of monthly metrics for the groups

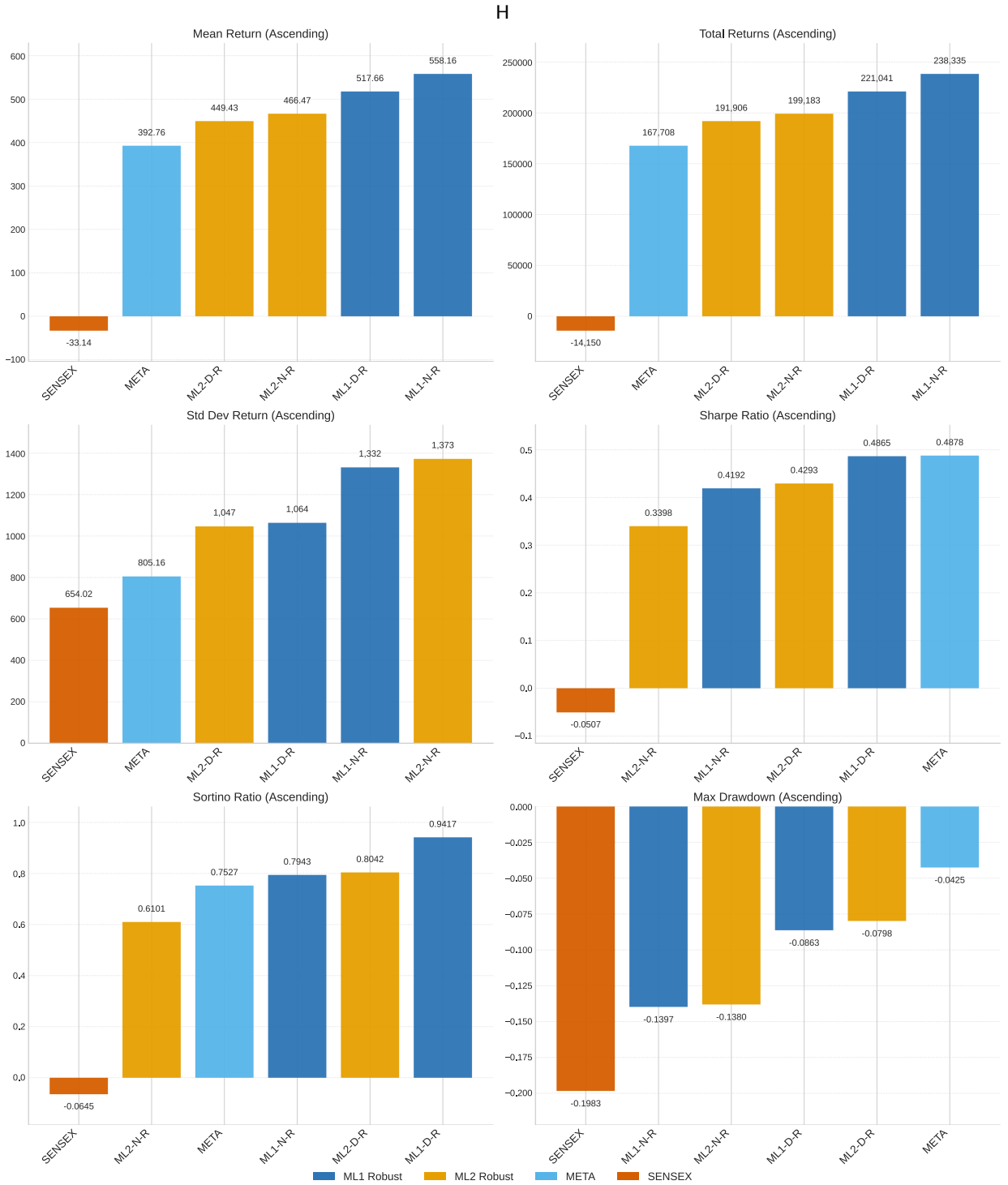


Figure 2: Legend: Robust metrics

References

- Brogaard, J., Zareei, A., 2023. Machine learning and the stock market. *Journal of Financial and Quantitative Analysis* 58, 1431–1472. doi:10.1017/S0022109022001120.
- Chen, W., Zhang, H., Mehlawat, M., Jia, L., 2021. Mean–variance portfolio optimization using machine learning-based stock price prediction. *Applied Soft Computing* 100, 106943. doi:10.1016/j.asoc.2020.106943.
- Engels, M., 2004. Portfolio Optimization: Beyond Markowitz. Master's thesis. Leiden University.
- Ma, Y., Han, R., Wang, W., 2021. Portfolio optimization with return prediction using deep learning and machine learning. *Expert Systems with Applications* 165, 113973. doi:10.1016/j.eswa.2020.113973.
- Mishra, S., Padhy, S., 2019. An efficient portfolio construction model using stock price predicted by support vector regression. *The North American Journal of Economics and Finance* 50, 101027. doi:10.1016/j.najef.2019.101027.
- Mishra, S., Padhy, S., Mishra, S.N., Misra, S.N., 2021. A novel lasso–tlbo–svr hybrid model for an efficient portfolio construction. *The North American Journal of Economics and Finance* 55, 101350. doi:10.1016/j.najef.2020.101350.
- Paiva, F.D., Cardoso, R.T., Hanaoka, G., Duarte, W., 2019. Decision-making for financial trading: A fusion approach of machine learning and portfolio selection. *Expert Systems with Applications* 115, 635–655. doi:10.1016/j.eswa.2018.08.003.
- Rosadi, D., Setiawan, E.P., Templ, M., Filzmoser, P., 2020. Robust covariance estimators for mean-variance portfolio optimization with transaction lots. *Operations Research Perspectives* 7, 100154. doi:10.1016/j.orp.2020.100154.
- Ruppert, D., 2004. *Statistics and finance: An introduction*. volume 27. Springer.
- Sharma, M., Shekhawat, H., 2022. Portfolio optimization and return prediction by integrating modified deep belief network and recurrent neural network. *Knowledge-Based Systems* 250, 109024. doi:10.1016/j.knosys.2022.109024.
- Shwartz-Ziv, R., Armon, A., 2022. Tabular data: Deep learning is not all you need. *Information Fusion* 81, 84–90.
- Xidonas, P., Steuer, R.E., Hassapis, C., 2020. Robust portfolio optimization: a categorized bibliographic review. *Annals of Operations Research* 292, 533–552. doi:10.1007/s10479-020-03630-8.
- Zhou, Z., Song, Z., Ren, T., Yu, L., 2022. Two-stage portfolio optimization integrating optimal sharpe ratio measure and ensemble learning. *IEEE Access* 11, 1654–1670. doi:10.1109/ACCESS.2022.3233260.